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Machine Learning

Report : Data Cleaning and Different Model Implementation for Student Depression Status.

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1. Introduction

This project aims to implement and compare multiple classification algorithms to understand their behavior and performance. The dataset used simulates student depression data, and while the focus is not to identify mental health patterns, it serves as an educational tool to apply machine learning techniques. This report documents the data-cleaning process and the implementation of a Decision Tree model.

2. Data Cleaning

To ensure reliable and interpretable results, the dataset underwent a thorough cleaning process. Below are the steps followed:

2.1 Data Overview

The dataset, "Student Depression Dataset," contains demographic, academic, and lifestyle attributes of students. The target variable is `Depression_Status`, a binary indicator (Yes/No) of whether a student is experiencing depression.

2.2 Handling Missing Values

- The dataset was scanned for missing values in all columns.
- Rows with critical missing values were removed to maintain data integrity.

2.3 Feature Selection

Irrelevant columns were removed to streamline the dataset for modeling. These included:

- `Id` (unique identifier)
- `City` (geographical information)
- `Profession` (profession of parents or students)
- `Work Pressure` (removed due to uniform values across samples)
- `Job Satisfaction` (removed due to uniform values across samples)

2.4 Encoding Categorical Variables

Categorical features such as `Gender`, `Sleep Duration`, and `Dietary Habits` were encoded using one-hot encoding.

2.5 Normalization and Scaling

Continuous variables were standardized to ensure they contributed equally to the model's learning process.

2.6 Final Dataset

After preprocessing, the dataset included the following **18 features** and **[Y] samples**, ready for model training:

Age, Academic Pressure, CGPA, Study Satisfaction, Work/Study Hours, Financial Stress, Depression, Gender_Male, Sleep Duration_7-8 hours, Sleep Duration_Less than 5 hours, Sleep Duration_More than 8 hours, Sleep Duration_Others, Dietary Habits_Moderate, Dietary Habits_Others, Dietary Habits_Unhealthy, Family History of Mental Illness_Yes, Have you ever had suicidal thoughts?_Yes.

3. Model Training using Different Algorithms

3.1 Decision Tree Model Implementation

Decision Trees are intuitive and powerful algorithms for classification tasks. The implementation of the Decision Tree model involved several key steps:

Data Splitting:

- The dataset was divided into training and testing subsets using an 80-20 split, with stratified sampling to retain the original class distribution.

Model Training:

- A Decision Tree Classifier was initialized with default hyperparameters.
- The model was trained on the training subset to learn patterns and relationships between the features and the target variable.

Hyperparameter Tuning:

- Key hyperparameters such as `max_depth` and `min_samples_split` can be fine-tuned. However, initial results are reported using default settings.

Performance Evaluation:

The trained model was evaluated on the test subset using the following metrics:

- **Accuracy:** 76%
This means the Decision Tree model correctly predicted whether a student was depressed or not 76% of the time.
- **Confusion Matrix:** Visual representation of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).
- **Classification Report:** Includes precision, recall, and F1-score for both classes.
 - **Precision:**

- 73% (No): Precision tells us how many of the students predicted to be not depressed actually were. For "No" (not depressed), 73% were correctly predicted.
 - 81% (Yes): Precision tells us how many of the students predicted to be depressed actually were. For "Yes" (depressed), 81% were correctly predicted.
- **Recall:**
 - 73% (No): Recall shows how many of the students who were actually not depressed (No) were correctly identified. For "No," the model identified 73% of actual non-depressed students.
 - 81% (Yes): Recall shows how many of the students who were actually depressed (Yes) were correctly identified. For "Yes," the model identified 81% of students who were actually depressed.
- **F1-Score:**
 - 73% (No): The F1-Score represents a balance between precision and recall. For "No," the balance is 73%.
 - 81% (Yes): The F1-Score represents a balance between precision and recall. For "Yes," the balance is 81%.
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Insights:

- The Decision Tree provided interpretability, with feature importance highlighting key predictors of depression.
- The model's performance could be enhanced through hyperparameter tuning and feature engineering.

3.2 Naive Bayes Model Implementation

Model Overview:

The Gaussian Naive Bayes model was applied to the dataset for classification. The model was trained using the training dataset and tested on the test dataset.

Key Performance Metrics:

- **Accuracy:** 83.28%
The Naive Bayes model correctly predicted the depression status of students 83.28% of the time.
- **Precision:** 88.57%
88.57% of the students predicted to be depressed were actually depressed.
- **Recall:** 82.03%
Out of all the students who were truly depressed, 82.03% were identified by the model.
- **F1-Score:** 85.17%
The F1-Score reflects a balance between precision and recall, indicating a strong

performance with 85.17% balance between avoiding false positives and correctly identifying true positives.

Confusion Matrix Insights:

- **True Positives (TP):** 3267 positive cases (Class 1) correctly identified.
- **True Negatives (TN):** 2313 negative cases (Class 0) correctly predicted.
- **False Positives (FP):** 591 cases incorrectly predicted as positive.
- **False Negatives (FN):** 419 positive cases missed and predicted as negative.

Analysis:

- **Overall Performance:**
Naive Bayes performed well with an accuracy of 83.28%, demonstrating a solid classification performance.
- **Precision and Recall Balance:**
The model showed a high precision of 88.57%, indicating better performance in minimizing false positives compared to recall (82.03%).
- **Impact of False Positives:**
Despite the high precision, there are some false positives, reflecting areas where the model predicted students to be depressed when they were not. This could be an area for further improvement.

3.3 Logistic Regression Model Implementation

Model Overview:

Logistic Regression was applied to the dataset using standardized data. The model's performance was evaluated based on accuracy, precision, recall, F1-score, and the confusion matrix.

Key Performance Metrics:

- **Accuracy:** 84.77%
Logistic Regression correctly predicted the depression status of students 84.77% of the time.
- **Precision:** 84.72%
84.72% of the students predicted to be depressed were actually depressed.
- **Recall:** 84.77%
84.77% of the students who were actually depressed were correctly identified by the model.
- **F1-Score:** 84.71%
The F1-Score of 84.71% indicates a strong balance between precision and recall, ensuring minimal false positives and false negatives.

Confusion Matrix Insights:

- **True Positives (TP):** 3267 positive cases (Class 1) correctly identified.
- **True Negatives (TN):** 2313 negative cases (Class 0) correctly predicted.
- **False Positives (FP):** 531 cases incorrectly predicted as positive.
- **False Negatives (FN):** 766 positive cases missed and predicted as negative.

Analysis:

- **Overall Performance:**
Logistic Regression provided strong, balanced performance with an accuracy of 84.77%, reflecting a high level of predictive accuracy.
- **Precision and Recall Balance:**
The model's precision (84.72%) and recall (84.77%) are closely aligned, indicating good performance in both identifying depressed students and minimizing false positives.
- **Impact of False Negatives:**
Despite the strong performance, the model missed 766 positive cases (FN), suggesting there is room for improvement, particularly in reducing false negatives.

3.4 Support Vector Machine (SVM) Implementation

Model Overview:

The Support Vector Machine (SVM) model was trained on the dataset and tested on a separate test set for classification.

Key Performance Metrics:

- **Accuracy:** 84.59%
The SVM model correctly predicted the depression status of students 84.59% of the time.
- **Precision:** 85.08%
85.08% of the students predicted to be depressed were actually depressed.
- **Recall:** 89.35%
89.35% of the students who were actually depressed were correctly identified by the model.
- **F1-Score:** 87.16%
The F1-Score of 87.16% represents an excellent balance between precision and recall, with a higher recall reflecting the model's ability to correctly identify more depressed students.

Confusion Matrix Insights:

- **True Positives (TP):** 3267 positive cases (Class 1) correctly identified.
- **True Negatives (TN):** 2313 negative cases (Class 0) correctly predicted.
- **False Positives (FP):** 810 cases incorrectly predicted as positive.
- **False Negatives (FN):** 95 positive cases missed.

Analysis:

- **Overall Performance:**

The SVM model was highly accurate (84.59%) with excellent recall (89.35%) for positive cases.

- **Precision and Recall Balance:**

The model showed good precision (85.08%) and excellent recall, but false positives (810) suggest that there is room for improvement in the classification of negative cases.

Conclusion

The models evaluated in this study—**Decision Tree**, **Naive Bayes**, **Logistic Regression**, and **Support Vector Machine (SVM)**—all performed reasonably well, but each showed distinct strengths and weaknesses in predicting depression status among students.

Accuracy Comparison:

- The **Logistic Regression** model achieved the highest accuracy of 84.77%, closely followed by the **SVM** model at 84.59%. The **Naive Bayes** model had a slightly lower accuracy of 76.65%, while the **Decision Tree** model had an accuracy of 77.24%. This indicates that, while the Decision Tree and Naive Bayes models were still reasonably accurate, the Logistic Regression and SVM models outperformed them.

Precision and Recall:

- The **SVM** model exhibited the highest recall (89.35%), meaning it identified the highest proportion of actual positive cases (depressed students). This suggests that the SVM model was particularly good at detecting students who were depressed, though it also had a relatively high number of false positives (810 cases), which indicates room for improvement in reducing false alarms.
- **Logistic Regression** and **SVM** had very closely aligned precision and recall values, showing balanced performance between correctly identifying depressed students and minimizing false positives. The **Logistic Regression** model's precision (84.72%) and recall (84.77%) were very similar, reflecting a solid overall performance.
- On the other hand, the **Decision Tree** and **Naive Bayes** models had lower recall values, suggesting that they missed a higher number of actual depressed students compared to SVM and Logistic Regression. This is evident as their recall values were in the 70% range, which indicates that these models were less effective at identifying all the depressed students.

F1-Score:

- The **SVM** model had the best F1-Score of 87.16%, followed closely by **Logistic Regression** at 84.70%. The **Decision Tree** and **Naive Bayes** models had lower

F1-Scores, with **Decision Tree** at 73.68% and **Naive Bayes** at 79.74%. The higher F1-Score of SVM and Logistic Regression suggests they achieved a better balance between precision and recall, with SVM providing the best overall balance.

Confusion Matrix Insights:

- The **SVM** model, with its superior recall, correctly identified a higher number of true positives (depressed students) compared to the other models. However, it also had a higher number of false positives, which indicates that while it was effective at identifying depressed students, it occasionally misclassified non-depressed students as depressed.
- The **Naive Bayes** model had the fewest false positives (908), but its lower recall suggests it missed a considerable number of actual depressed students. This indicates that while Naive Bayes was good at minimizing false positives, it was less effective at identifying all the positive cases.

Summary:

- The **SVM** model demonstrated the best recall and F1-Score, making it the most reliable for identifying students with depression, though it could benefit from minimizing false positives.
- **Logistic Regression** provided a solid, well-balanced performance, performing just slightly behind SVM, with the advantage of a well-aligned precision and recall.
- **Decision Tree** and **Naive Bayes** models were less effective in terms of recall and accuracy, with **Naive Bayes** showing a stronger focus on minimizing false positives at the expense of recall.

Overall, the **SVM** and **Logistic Regression** models stand out as the most robust for depression detection, with **SVM** excelling in recall and **Logistic Regression** offering well-rounded performance. Further tuning of the **SVM** model could potentially address the issue of false positives and improve its overall performance.